

Deepak Mallampati

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Kent, OH - Open to Relocation (San Francisco, CA)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in production LLM systems for healthcare - RAG pipelines, agentic workflows, and evaluation frameworks that make agents measurably better. At Progress Solutions, built clinical knowledge retrieval systems (~35% retrieval precision improvement), multi-step healthcare agents with structured reasoning and grounding rules (~25% stability improvement), and evaluation pipelines tracking retrieval alignment, hallucination rate, and agent stability (90%+ grounded alignment). MS, Kent State University. Comfortable owning end-to-end delivery: data ingestion, prompt engineering, agent orchestration, evaluation design, and Docker-based production deployment in HIPAA-conscious environments.

CORE COMPETENCIES

LLM Application Development • RAG & Semantic Retrieval • Agentic Workflows • Evaluation Pipelines & Guardrails • Prompt Engineering & Grounding • Python & FastAPI • ML Pipelines (scikit-learn, XGBoost, PyTorch) • Healthcare AI & HIPAA • Docker & CI/CD • LangChain & LlamaIndex

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Built **Retrieval-Augmented Generation (RAG)** systems for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%. Retrieval-grounded response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by 15-20% on high-risk categories while holding precision >90% via class weighting, stratified sampling, and threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above 98% and cut downstream model instability by ~40%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana**; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive environment.

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization system** across 5,000+ sessions; reduced manual review time by **60-70%** and achieved **~85% alignment** with human-curated highlights.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs **~30%** and raised contextual accuracy **>85%**. Deployed as a Flask-based API for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, RAG, JSON Schema, Vector Search

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped · Chrome Web Store

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/local). Multi-section capture pipeline, seven-day translation cache, per-domain selector configs for **29 manga/webtoon sites**, narrowed host permissions.

TypeScript, React, Manifest V3, multi-provider AI APIs, Chrome Web Store

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript app; coordinated multimodal APIs (Perplexity Sonar + Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

FastAPI, React, TypeScript, Vite, Tailwind, Perplexity API, Replicate

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with unified **YOLOv8** model; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping and NMS tuning for stable real-time operation via OpenCV.

YOLOv8, OpenCV, Python, LabelImg, PyTorch

E-Farming Digital Marketplace founder · full-stack developer

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80-120 active users** in phase one; cart, reviews, and community blogging features.

PHP, MySQL, Bootstrap, full-stack marketplace

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, Retrieval-Augmented Generation (RAG), agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems